COMED KARES
Community Centric Education

ERA
FOUNDATION

Student Performance Prediction

Data Science and AI ML

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Agenda:

- Introduction
- SMART Objectives
- 5W1H Analysis and SCAMPER Ideation
- Literature Insights (Short)
- Proposed Solution
- System Overview (Basic Diagram)
- Gantt Chart
- Expected Outcome

Problem Statement

“We aim to develop a Machine Learning–based Student Performance Prediction System to solve the problem of late identification of academically at-risk students for educational institutions and instructors.”

Educational institutions often evaluate student performance only after final examinations, making it difficult to support struggling students in time. The proposed system uses machine learning techniques to analyze academic data such as attendance, marks, and assignments to predict student performance early. This helps teachers and institutions take timely actions to improve learning outcomes and reduce failure rates.

- ❑ Predict student academic performance at an early stage
- ❑ Help teachers take preventive academic actions
- ❑ Improve overall student success rate
- ❑ Reduce dropout and failure rates

Introduction

Problem Area:

Who Faces This Problem?

- Students who struggle academically and are not identified early
- Teachers who find it difficult to monitor large classrooms
- Educational institutions aiming to improve results and reduce failures
- Students who struggle academically and are not identified early

Where Does It Occur?

It commonly occurs in schools, colleges, and online learning environments where large volumes of student data are generated but not effectively analyzed. This issue is important because late identification of poor performance leads to higher failure rates and missed opportunities for timely academic support. Addressing this problem through predictive systems can significantly improve learning outcomes and overall educational quality

Why Is It Important?

- Late identification of weak students leads to poor academic outcomes
- Increases failure rates and dropout risks

Objectives (SMART)

1. **Objective : Specific**

To develop a machine learning-based system that predicts student academic performance using factors such as attendance, marks, and study behavior.

2. **Objective : Measurable**

To achieve high prediction accuracy (e.g., above 85–90%) by evaluating the model using metrics like accuracy, precision, and recall.

3. **Objective : Achievable**

To implement the system using available datasets and standard machine learning algorithms within the scope of academic resources and tools.

4. **Objective : Relevant**

To address the need for early identification of at-risk students and improve academic outcomes in educational institutions.

5. **Objective : Time Bound**

To design, develop, and evaluate the system within the defined academic project timeline (e.g., one semester).

5W1H Analysis

Category	Description
WHO	Teachers, faculty, academic administrators (primary users); parents, counselors, and institution management (secondary stakeholders); students (beneficiaries)
WHY	To identify academically weak students early, improve success rates, reduce dropout rates, and enable data-driven academic decisions
HOW	Using Machine Learning models (e.g., Random Forest, Decision Tree, Logistic Regression) applied on student data like attendance, marks, and behavior through a data-processing pipeline
WHAT	A Student Performance Prediction System that analyzes academic and behavioral data to predict future student outcomes (high, average, at-risk)
WHERE	Educational institutions such as schools, colleges, universities; can be deployed as a web-based or cloud-based system integrated with institutional platforms
WHEN	During the academic term (before final exams), enabling early prediction and timely intervention instead of post-result evaluation

SCAMPER Ideation

1. **Substitute :** Replace manual monitoring by teachers with an AI chatbot that automatically interacts with students and provides performance insights.

2. **Adapt:** Adapt chatbot systems used in customer support to the education domain for answering academic queries and tracking student performance.

3. **Put to Another Use:** Use the prediction system not only for performance analysis but also for career guidance and personalized learning recommendations.

4. **Eliminate:** Remove manual data analysis and reduce dependency on traditional evaluation methods by automating prediction and monitoring processes.

Literature Insights

Search words used:

Student Performance Prediction, Machine Learning in Education, Educational Data Mining, Learning Analytics, Academic Performance Prediction, AI in Education.

Papers selected:

“Student Performance Prediction Using Machine Learning” – E. Ahmed et al. (2024)

“Machine Learning-based Academic Performance Prediction” – W. Ahmed et al. (2025)

“Evaluation of ML Models for Student Performance” – Sandeepa & Mohottala (2025)

“Graph-Based Ensemble Learning for Student Prediction” – Wang et al. (2021)

“Neural Network-Based Student Performance Forecasting” – Junejo et al. (2024)

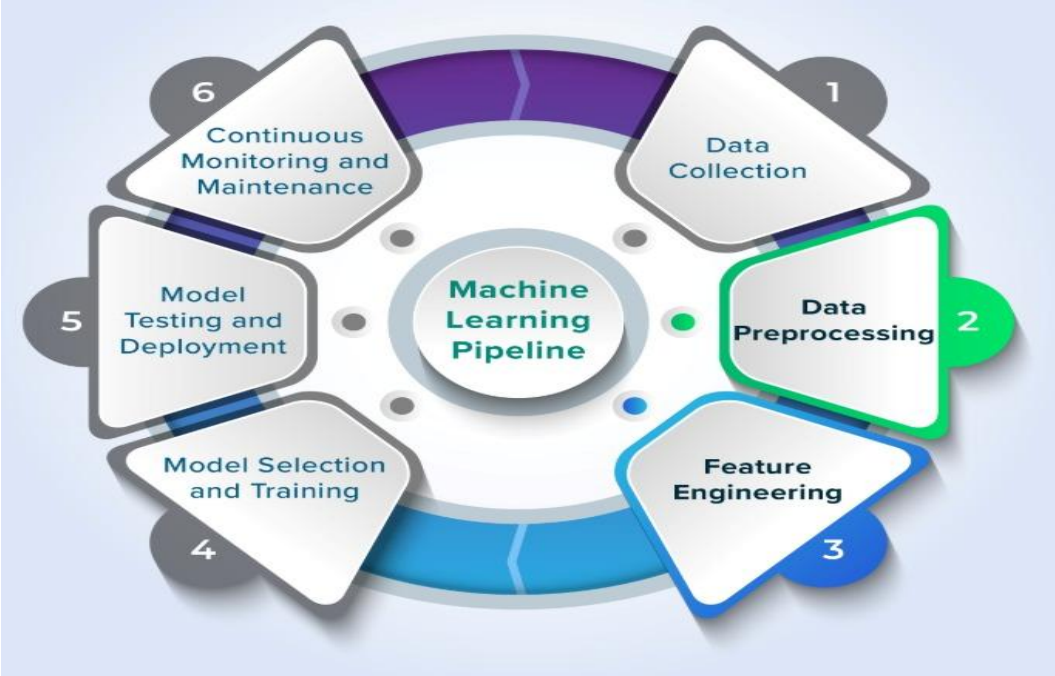
Gaps between existing solution and need:

Many systems are complex, less interpretable, and not easily adaptable to different educational environments. There is a need for a simpler, scalable, and more accurate predictive system that provides timely insights and supports proactive decision-making.

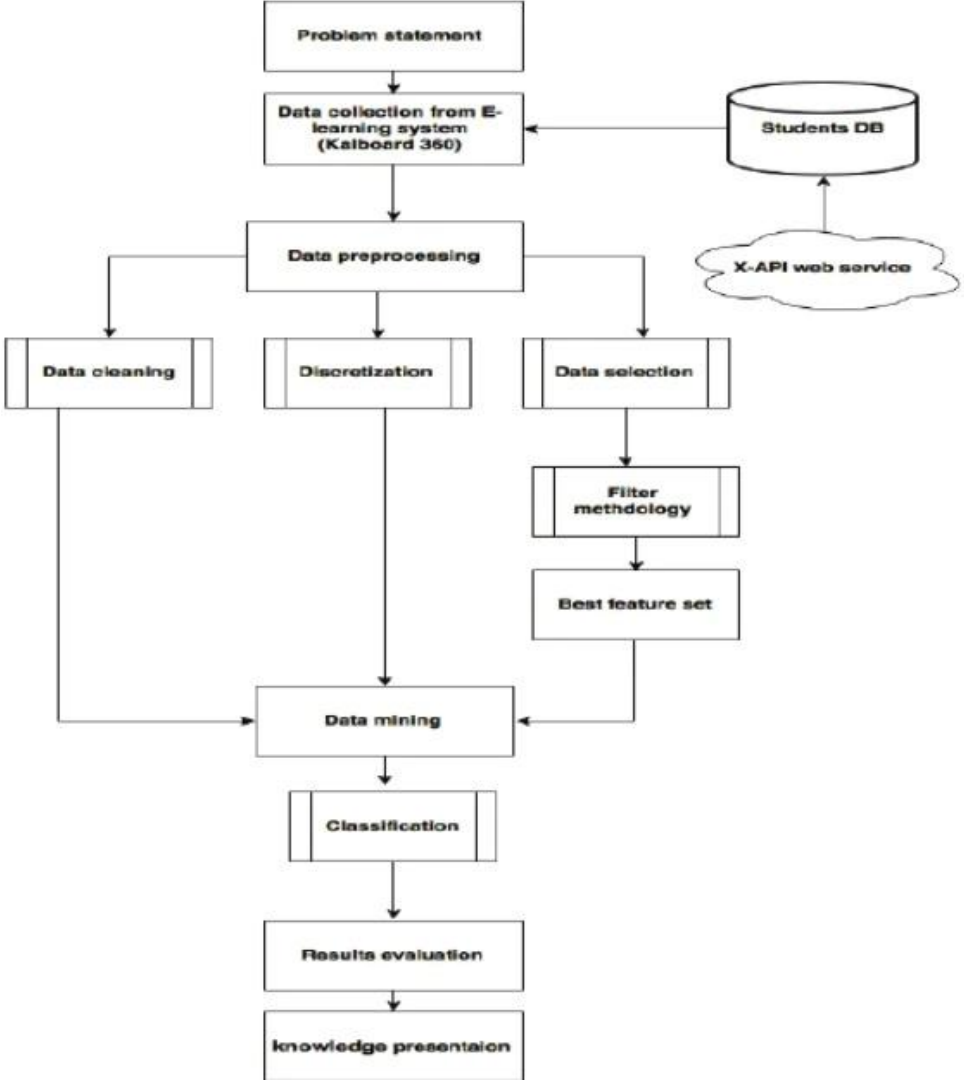
Proposed Solution

The proposed solution is a Machine Learning–based system that analyzes student data such as attendance, internal marks, assignment scores, and study behavior to predict academic performance in advance .It classifies students into categories such as high, average, and at-risk, enabling early identification of weak students. This allows teachers and institutions to take timely corrective actions, improve learning outcomes, and reduce failure rates.

- Uses Machine Learning models to predict student performance
- Analyzes data like attendance, marks, assignments, and behavior
- Classifies students into High / Medium / At-Risk categories



Installation & System Overview Configuration:



Project Details Final	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10	Week 11	Week 12	Week 13	Week 14	Week 15	Week 16
Problem definition & planning		■	■													
Data collection			■	■												
Data cleaning & Preprocessing			■	■	■											
Exploratory Data Analysis				■	■											
Feature Engineering					■	■	■									
Model Selection & Trainig								■	■	■	■					
Model Evaluation										■	■	■				
Hyperparameter Tuning											■	■	■			
Model Testing												■	■	■		
Deployment														■	■	
Final Report & Presentation																■

Expected Outcome

The system will successfully predict student academic performance using machine learning models based on input data such as attendance, marks, and study behavior. It will classify students into categories like High Performer, Average Performer, and At-Risk Student, enabling early identification of weak students. The system will also generate insights and visualizations to support teachers and institutions in making data-driven academic decisions.

Example:
Attendance: 90%
Study Hours: 4 hr/day
Internal Marks: 85
Predicted Result: High Performer